

LayerZero: Blockchain-Based Carbon Credit MRV System

Prof. Sadhana Kekan, Priyanshu Lamkhad, Om Raosaheb Kale, Ayush Rahul Mane, Gaurav Prasad Patil

Department of Information Technology, Progressive Education Society's, Modern College of Engineering, Pune, Maharashtra, India

ARTICLE INFO

Article History:

Accepted : 20 May 2026

Published: 19 June 2026

Publication Issue

Volume 12, Issue 3

May-June-2026

Page Number

815-821

ABSTRACT

The credibility of voluntary carbon markets depends heavily on the reliability of the systems used to measure, report, and verify carbon sequestration claims. Conventional infrastructure for monitoring, reporting, and verification is predominantly manual, fragmented, and vulnerable to data manipulation, leading to inflated credit claims and double-counting. We design and implement LayerZero, a full-stack blockchain-based Carbon credit validation and tokenization platform. The system integrates guided project submission workflows, geospatial plantation boundary geofencing via Leaflet, field-level biomass carbon quantification based on adapted IPCC Tier 1 equations, tamper-evident image storage via Cloudinary, and automatic ERC-20 token minting on the Ethereum Sepolia testnet. A role-stratified governance model distributes system access among project owners, certified validators, and platform administrators. All verified carbon values are stored and publicly readable on-chain, creating a decentralized and auditable record. The system also supports annual monitoring cycles with incremental token minting, automated PDF certificate generation, and an integrated peer-to-peer token marketplace. System evaluations confirm that LayerZero successfully eliminates double-counting through token burning, reduces administrative delays, and provides a cost-effective digital accounting model for afforestation and reforestation projects.

Keywords : Blockchain, Carbon Credit, MRV System, ERC-20, Ethereum Sepolia, Smart Contract, Carbon Sequestration, Decentralized Application, PostgreSQL, Next.js, IPCC Biomass.

I. INTRODUCTION

Climate change remains one of the most significant challenges of our time, driving the need for effective market-based solutions to reduce global greenhouse gas emissions [4]. Carbon credits serve as a key mechanism in this effort, allowing entities to offset

their emissions by purchasing units of carbon dioxide sequestered by environmental projects. One carbon credit corresponds to one metric ton of carbon dioxide removed from or prevented from entering the atmosphere. As commitment to the Paris Agreement has grown, the voluntary carbon market has expanded rapidly, reaching billions of dollars in traded value

and projecting significant growth over the coming decade [4]. However, the credibility of the carbon credit market is undermined by inefficiencies in monitoring, reporting, and verifying the actual carbon captured by these projects.

Traditional systems for monitoring, reporting, and verification (MRV) rely on manual processes that are slow, costly, and lack transparency [7]. Project owners must submit paper-based documentation, which is then reviewed by third-party auditors who perform physical field visits. The entire process can take months or even years, introducing delays and significant administrative costs. Centralized registries operate in silos, creating a double-counting risk where the same carbon sequestration project is registered and sold under different registries [4], [7]. This lack of coordination and public auditability compromises market trust and slows down environmental financing. Blockchain technology offers a decentralized ledger system that can address these administrative issues [1]. By storing records on a distributed network, the history of carbon credit projects becomes computationally immutable and transparent. Smart contracts can automate the validation and credit issuance workflow, ensuring tokens are minted only when verification parameters are met [2]. Integrating geospatial boundary mapping, scientific biomass equations, and secure on-chain tokenization establishes a reliable framework where each credit is linked to verifiable ecological data.

This work introduces LayerZero, a blockchain-based carbon credit validation and tokenization platform operating on the Ethereum Sepolia network. The system coordinates the project submission, validator assignment, carbon quantification, token minting, annual monitoring, and token marketplace operations. By combining off-chain relational database storage via PostgreSQL and Prisma ORM with on-chain Ethereum ERC-20 smart contracts [2], LayerZero provides a digital accounting system that eliminates double-counting, improves verification speed, and ensures public auditability of offset claims.

II. LITERATURE REVIEW

The application of blockchain technology to carbon accounting and tokenization has been explored in several recent studies. Researchers have focused on addressing the opacity and manual overhead associated with conventional environmental registries. Woo et al. [5] proposed a blockchain-based digital MRV architecture for building energy performance monitoring, demonstrating that blockchain provides tamper-proof data logging and automated compliance checking. However, their system was limited to building energy data and did not address the complexities of ecological monitoring.

Khan and Ahmad [6] developed DCarbonX, a decentralized application for carbon market simulation. Their Ethereum-based prototype demonstrated token minting for carbon credits but lacked a scientific verification pipeline, relying on self-reported data without independent validation. Romijn et al. [7] advocated for transparent and open data in National Forest Inventories, emphasizing the need for standardized data formats and interoperable monitoring systems to support high-integrity carbon markets.

Bachina [8] explored efficient RESTful API development patterns using Node.js, Express, and PostgreSQL, providing architectural patterns that informed our backend design decisions. LayerZero combines these technologies with an Ethereum ERC-20 smart contract [2] to resolve the registry fragmentation and oracle problems identified in the literature, providing a transparent, end-to-end framework for digital carbon tracking. Table I provides a comparative analysis of these methodologies.

Table I. Comparative Analysis of Existing Systems

Sr. No.	System / Study	Methodology / Key Features	Year
1	Woo et al. [3]	Blockchain digital MRV architecture for building energy performance; smart contract audits.	2020
2	Khan & Ahmad [4]	DCarbonX decentralized simulation for Ethereum-based carbon token minting.	2022
3	Romijn et al. [5]	Standardized open-data protocols for forest monitoring and data validation.	2022
4	Bachina [6]	Separated backend pattern with Node.js, Express, and PostgreSQL for robust routing.	2023
5	Marr [7]	Conceptual framework for tracking emissions and voluntary credits via decentralized ledger.	2022
6	Khaqqi et al. [8]	P2P carbon emissions trading with reputation assessment on private blockchain.	2018
7	Abiodun et al.	Systematic review	2025

	[9]	of blockchain integration in carbon markets and smart oracle analysis.	
8	Chapron [10]	Cryptogovernance framework and decentralized verification safeguards for conservation.	2017

III. PROPOSED SYSTEM

The proposed system is designed as a web-based application that provides structured access to carbon credit MRV. It integrates a React-powered frontend with an Express.js service layer, enabling efficient data retrieval, search, and tokenization. The overall architecture follows a client-server model and is divided into three major components:

A. System Architecture

The LayerZero platform follows a three-tier architecture designed to balance secure data processing, user accessibility, and public auditability. The three tiers consist of a client web application, a RESTful backend server, and a dual-persistence database and blockchain layer. Figure 1 shows the overall system architecture, illustrating the connections between users, validators, administrators, the backend server, and the Ethereum Sepolia network.

Presentation Layer (Frontend): The presentation layer is built using Next.js 15 and React 19, which uses a component-based structure to make the code modular, scalable, and efficient. It handles features like drawing geofenced boundaries, filtering project criteria, and connecting browser wallets (MetaMask) asynchronously via Ethers.js.

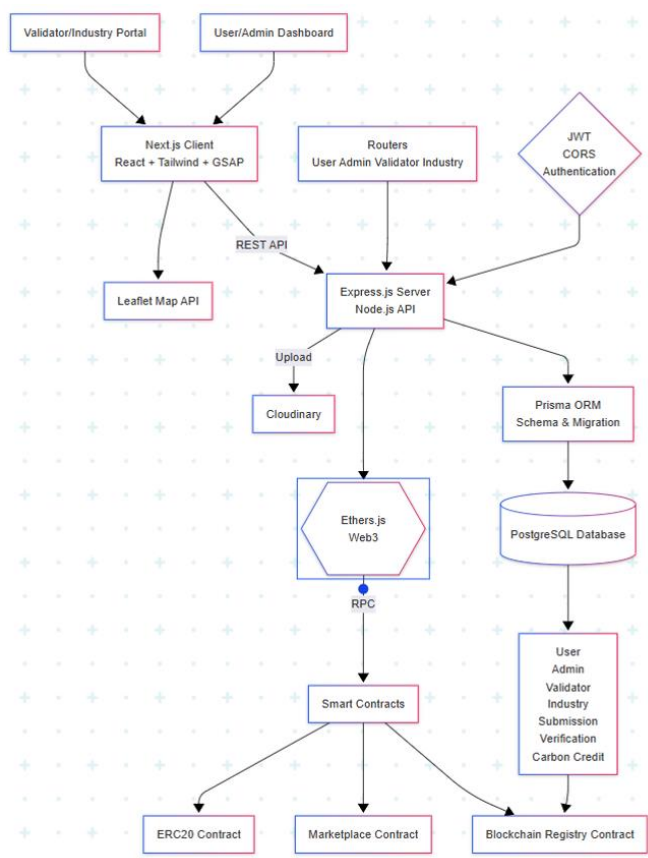


Fig. 1: Proposed System Architecture of LayerZero

Application Layer (Backend API): The application layer is built as a RESTful Express.js API. It handles request routing, role-based authorization middleware (JWT), file dispatch to Cloudinary, and smart proximity scheduling to assign nearest available validators based on coordinates.

Data Layer (Persistence): Consists of a relational PostgreSQL database accessed via Prisma ORM for structural data tracking (users, validation inputs), paired with an Ethereum Sepolia blockchain layer executing ERC-20 smart contract logic for immutable minting and retirement transactions.

B. Workflow

The proposed system presents integrated workflows for landowners and field validators within the platform. For landowners, the process begins with project submission, drawing boundary polygons, and uploading geotagged images. For validators, the workflow starts with proximity task assignment. Once

accepted, the validator conducts a field audit, records parameters (survival rate, canopy height), and generates an audited verification report, which triggers the backend contract minting service.

C. Technical Features

Key technical features include client-side boundary area calculations using Turf.js, automated proximity scheduler algorithms, Express.js stateless routing, Puppeteer-driven PDF generation for offset certificates, and on-chain ERC-20 token controls (mintToLandowner, retire, balanceOf).

IV. METHODOLOGY

The project is designed using modular principles and modern information retrieval methods. The development process is divided into project submission, validator assignment, carbon quantification, and token operations.

A. Project Submission and Geofencing

The workflow starts when a landowner registers a project. The submission form collects the physical address, manual GPS coordinates, and a boundary polygon drawn on the interactive Leaflet map. The user also uploads a geotagged site photograph, specifies the plantation date, inputs up to three tree species with their respective counts, and declares the claimed area. Turf.js calculates the area of the polygon drawn on the map, which the database records alongside the self-reported claim. This allows validators to detect discrepancies between declared areas and mapped geofenced zones [12].

B. Validator Assignment

Once a submission is created, the backend assignment engine schedules a certified field validator. The scheduler selects validators based on three criteria: geographic proximity (using the validator's registered base coordinates), current workload (with a limit of three active assignments), and an availability flag. The validator receives an assignment notification. Accepting the task changes the project status from

PENDING to INPROGRESS. If a validator does not accept the assignment within a specified deadline, the scheduler marks the assignment as EXPIRED and re-routes the project to another available validator.

C. Carbon Quantification Model

During the field audit, the validator evaluates the project site and inputs parameters into a three-part verification form. Section A records general site indicators (actual area, boundary match, vegetation density, soil condition, illegal activity, and pollution), Section B records biological parameters (survival rate, canopy height, plantation health, water condition, and planting method), and Section C displays the carbon values. The backend computes the carbon sequestration using adapted IPCC Tier 1 afforestation and reforestation biomass equations [9]. The calculations are performed as follows:

$$AGB = SR * CH * SDF(1)$$

where *AGB* represents the Above Ground Biomass in metric tons per hectare, *SR* is the survival rate percentage (expressed as a decimal), *CH* is the average canopy height in meters, and *SDF* is the species-specific biomass density factor (e.g. 0.5 for standard afforestation species).

$$BGB = AGB * 0.26(2)$$

where *BGB* is the Below Ground Biomass, calculated using the IPCC Tier 1 default root-to-shoot ratio of 0.26 for forest systems, representing root biomass carbon stocks [9], [10].

$$SC = Area * SOC * Y(3)$$

where *SC* is the Soil Carbon stock, computed by multiplying the project area by the soil organic carbon (*SOC*) accumulation rate (e.g. 1.2 tons of carbon per hectare per year) and the number of years (*Y*) since the plantation date.

$$TC = AGB + BGB + SC(4)$$

where *TC* is the Total Carbon stock accumulated within the project boundary, representing the sum of above-ground, below-ground, and soil carbon pools.

$$Annual\ CO2 = TC * 3.67(5)$$

where *Annual CO2* represents the equivalent carbon dioxide mass in tons. The conversion factor 3.67 reflects the molecular weight ratio of carbon dioxide to carbon (44/12). One carbon token represents one kilogram of carbon dioxide equivalent (*CO2e*), meaning that the number of tokens issued is equal to the *Annual CO2* value multiplied by 1,000.

D. Token Minting and On-Chain Storage

When a validator approves a verification report, the system updates the history status to APPROVED and initiates token minting. The Express.js backend initiates a transaction call to the deployed smart contract [12]. The call transfers the computed annual *CO2e* value (converted to units) to the landowner's public wallet address. The transaction hash returned by the Sepolia network is saved in the database, creating a link between the off-chain project parameters and the on-chain event.

E. Annual Monitoring and Re-Verification

Approved carbon projects are subject to annual re-verification. The scheduler generates a monitoring task each year, assigning a field validator to re-evaluate the site. The validator records current survival rates, canopy height, and vegetation health. The system recalculates the carbon sequestration for the current year. If the new carbon stock exceeds the previous year's value, the server mints additional tokens for the incremental difference. If the carbon stock has decreased, no new tokens are issued, and the deficit is recorded. This incremental model prevents token over-issuance and accounts for ecological changes over time.

F. Token Retirement and Offset Certification

Industrial partners purchase carbon tokens through the integrated marketplace to offset emissions. To claim an offset, the partner invokes the retire function on the smart contract, burning the tokens and removing them from circulation [12], [13]. The transaction requires a string argument explaining the

offset reason. Upon confirmation, the backend uses Puppeteer to generate a PDF certificate containing the company name, offset volume, project details, the retirement transaction hash, and a QR code resolving to the Etherscan transaction page.

V. RESULTS AND DISCUSSIONS

The LayerZero platform was evaluated using test cases representing afforestation projects in Maharashtra. Project sizes ranged from 0.5 to 10 hectares. Form submissions, Leaflet boundary polygon calculations, and image uploads completed without errors. The validator assignment engine correctly scheduled validators based on geographic coordinates. Calculated carbon values matched manual formula verifications. Smart contract operations (minting, retirement, and balance queries) executed successfully on the Ethereum Sepolia testnet, with transaction hashes recorded in the PostgreSQL database. Table II summarizes the functional test results.

A. Performance Evaluation

Table II. Summary of Functional Test Results

Test Scenario	Expected Outcome	Result
Project submission	Form saved, boundary computed via Turf.js	Pass
Validator assignment	Nearest validator assigned based on coordinates	Pass
Carbon calculation	Biomass computed using IPCC Tier 1 equations	Pass
ERC-20 token minting	Tokens minted to owner Sepolia wallet; tx saved	Pass
PDF certificate gen	Puppeteer PDF generated with Etherscan QR	Pass
Annual monitoring	Incremental tokens minted based on growth	Pass

B. Security and Data Integrity

The dual-persistence design helps protect data integrity. Relational data is stored in the PostgreSQL database, while token transactions are secured by the Ethereum Sepolia network. If the backend database is compromised, the transaction history remains verifiable on the blockchain. JWT-based authentication and route-specific middleware prevent unauthorized access. Geotagged evidence images are uploaded directly to Cloudinary and linked to submissions, reducing the risk of document alteration. Smart contract access controls ensure that only the server's authorized wallet can execute the mintToLandowner function.

C. Limitations

The platform was evaluated using the Ethereum Sepolia testnet, which does not incur gas costs. Deploying to the Ethereum mainnet would require project owners to pay transaction fees, which could limit viability for small landowners. Implementing Layer 2 scaling networks (such as Polygon or Arbitrum) could help reduce these costs. The carbon quantification model uses IPCC Tier 1 defaults; integrating species-specific density databases would improve calculation accuracy. The system also relies on manual field audits; incorporating satellite remote sensing and drone imagery could automate boundary verification and canopy height measurements.

VI. CONCLUSION AND FUTURE WORK

This paper presented LayerZero, a blockchain-based carbon credit monitoring, reporting, and verification platform. The system coordinates the credit lifecycle from project submission and boundary geofencing to field auditing, carbon quantification, and token retirement. By combining relational database storage with smart contracts, LayerZero helps prevent double-counting and provides an auditable history for each carbon credit. The incremental monitoring model adjusts token issuance based on yearly biomass growth, offering a structured approach for forest

conservation projects. Future development will focus on integrating satellite boundary validation, expanding the species database, and deploying the platform on a Layer 2 network to reduce transaction costs.

Practices in Engineering & Management (IIPeM), Singapore, Singapore, 2025, pp. 1-6, doi: 10.1109/IIPeM65914.2025.11548421

References

- [1]. S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," 2008.
- [2]. V. Buterin, "A Next-Generation Smart Contract and Decentralized Application Platform," Ethereum White Paper, 2013.
- [3]. Intergovernmental Panel on Climate Change (IPCC), "Good Practice Guidance for Land Use, Land-Use Change and Forestry," Institute for Global Environmental Strategies, Hayama, Japan, 2003.
- [4]. Verra, "VM0047: Afforestation, Reforestation and Revegetation Methodology," Washington D.C., USA, Sep. 2023.
- [5]. J. Woo, S. Park, L. Kim, and M. Choi, "A New Blockchain Digital MRV Architecture for Existing Building Energy Performance," ResearchGate, Oct. 2020.
- [6]. N. Khan and T. Ahmad, "DCarbonX Decentralised Application: Carbon Market Case Study," arXiv preprint arXiv:2203.09508, Mar. 2022.
- [7]. E. Romijn, A. Herold, M. Angelsen, L. Verchot, B. Murdiyarso, and K. Martius, "Towards transparent and open data in National Forest Inventories," ResearchGate, Oct. 2022.
- [8]. B. Bachina, "Efficient RESTful API Development with NodeJS, Express, and PostgreSQL," European Journal of Advances in Engineering and Technology, vol. 10, no. 6, pp. 26-37, 2023.
- [9]. S. Mishra, "AI Automation Techniques for Optimized Scheduling and Resource Allocation in Genomics Data Pipelines," 2025 International Conference on Intelligent & Innovative